Example 9.11: The function $f(x) = \sin x$ is Lipschitz on $\mathbb{R}$ with constant $L = 1$ (which means that the Lipschitz constant is *at most* 1). Indeed, we have

$$\begin{aligned} |\sin x_1 - \sin x_2| &= 2\left|\sin\frac{x_1 - x_2}{2}\right|\left|\cos\frac{x_1 + x_2}{2}\right| \\ &\leq 2\left|\sin\frac{x_1 - x_2}{2}\right| \\ &\leq |x_1 - x_2|. \end{aligned}$$

Proposition 9.9: If f is Lipschitz on I, then it is also continuous on I.

Proof. Fix some point $x_0 \in I$. Since f is Lipschitz, there exists some $L \geq 0$ such that

$$|f(x_0) - f(x)| \leq L|x_0 - x|, \qquad \forall x \in I.$$

We may assume $L > 0$: if the inequality holds with some L, it also holds with any larger constant. Let $\varepsilon > 0$ and define $\delta = \frac{\varepsilon}{L}$. Then whenever $|x_0 - x| < \delta$ we have

$$|f(x_0) - f(x)| \leq L|x_0 - x| < L\delta = \varepsilon$$

which proves that f is continuous at x_0. Since x_0 was arbitrary, f is continuous on I. □

Uniform continuity

A function $f : \mathbb{R} \to \mathbb{R}$ is **uniformly continuous on an interval** $I \subseteq \mathbb{R}$ if for any $\varepsilon > 0$, there exists $\delta = \delta(\varepsilon) > 0$ such that

$$\forall x_1, x_2 \in I, \qquad |x_1 - x_2| < \delta \quad \Rightarrow \quad |f(x_1) - f(x_2)| < \varepsilon.$$

The crucial aspect in the above definition is that the same $\delta(\varepsilon)$ works for the *entire interval* I, not just at a single point. Recall that the definition of continuity held only at a given point x_0!

Corollary 9.10: If f is Lipschitz on I, then it is uniformly continuous on I.

Proof. In the proof of Proposition 9.9, $\delta = \frac{\varepsilon}{L}$ does not depend on the point x_0. That is, for every $\varepsilon > 0$ and all $x_1, x_2 \in I$ with $|x_1 - x_2| < \delta$, we have $|f(x_1) - f(x_2)| \leq L|x_1 - x_2| < L\delta = \varepsilon$. □

Example 9.12: The function $f(x) = \frac{1}{x}$ is *not* uniformly continuous on $(0, 1)$: as $x \to 0^+$, compare, for instance x and $\sqrt{x}$:

$$\lim_{x\to 0^+} |x - \sqrt{x}| = 0$$

while

$$\lim_{x\to 0^+} |f(x) - f(\sqrt{x})| = \lim_{x\to 0^+}\left|\frac{1}{x} - \frac{1}{\sqrt{x}}\right| = \lim_{x\to 0^+}\left|\frac{1 - \sqrt{x}}{x}\right| = +\infty.$$

So points that become arbitrarily close on the x-axis have images that become arbitrarily far apart on the y-axis.

However, $f(x) = \frac{1}{x}$ is uniformly continuous on any interval at a positive distance from 0. For instance, consider $I = [a, +\infty)$ where $a > 0$. Then

$$|f(x_1) - f(x_2)| = \left|\frac{1}{x_1} - \frac{1}{x_2}\right| = \frac{|x_1 - x_2|}{x_1 x_2} \leq \frac{1}{a^2}|x_1 - x_2|.$$

Hence for any $\varepsilon > 0$ we can take $\delta = a^2\varepsilon$ to get that if $|x_1 - x_2| < \delta$ then $|f(x_1) - f(x_2)| \leq \frac{1}{a^2}|x_1 - x_2| < \frac{1}{a^2}\delta = \frac{1}{a^2}a^2\varepsilon = \varepsilon$.

Example 9.13: The function $f(x) = \sin(\frac{1}{x})$ is also *not* uniformly continuous on $(0, 1)$. Indeed, as $x \to 0^+$ there are points that are arbitrarily close on the x-axis, while their distance on the y-axis is 2 (we're thinking here about points where the sine function achieves the values ± 1).

Theorem 9.11 (Heine-Cantor Theorem)**:** If f is continuous on a closed interval $I = [a, b]$, where $a < b$ are real numbers, then it is uniformly continuous there.

Proof. We prove *by contradiction*. If the claim is not true, then f is not uniformly continuous on I. That is, $\exists \varepsilon > 0$ such that $\forall \delta > 0$, $\exists x', x'' \in I$, such that $|x' - x''| < \delta$ and $|f(x') - f(x'')| \geq \varepsilon$.

So ε is fixed, but we can choose δ. Let $\delta = \frac{1}{n}$, $n \in \mathbb{N}_+$. Then for each n there exist $x'_n, x''_n \in I$ such that

$$|x'_n - x''_n| < \frac{1}{n} \qquad \text{and} \qquad |f(x'_n) - f(x''_n)| \geq \varepsilon.$$

We immediately observe that

$$\lim_{n\to\infty} |x'_n - x''_n| = 0,$$

while $|f(x'_n) - f(x''_n)| \geq \varepsilon$ for all n, so this difference cannot converge to 0. Both sequences $\{x'_n\}_{n\in\mathbb{N}_+}$ and $\{x''_n\}_{n\in\mathbb{N}_+}$ are bounded, so the Bolzano-Weierstrass theorem (Chapter 5) applies.

Let's start with the sequence $\{x'_n\}_{n\in\mathbb{N}_+}$: it is bounded, and therefore there is a convergent subsequence $\{x'_{n_k}\}_{k\in\mathbb{N}_+}$. Call its limit $\overline{x}$:

$$\lim_{k\to\infty} x'_{n_k} = \overline{x}.$$

Since $x'_n \in [a, b]$ for all n, Proposition 5.3 tells us that the limit point $\overline{x}$ must also belong to $[a, b]$.

Now we consider the sequence $\{x''_n\}_{n\in\mathbb{N}_+}$: it is also bounded, but we need to be cautious about working with it. We know that $|x'_n - x''_n| < \frac{1}{n}$, where it is important that both elements have the same index n. For the first sequence we already took a subsequence with index $\{n_k\}_{k\in\mathbb{N}_+}$. This forces us to only consider the elements of $\{x''_n\}_{n\in\mathbb{N}_+}$ that also have the indices $\{n_k\}_{k\in\mathbb{N}_+}$. Hence we should start with the subsequence $\{x''_{n_k}\}_{k\in\mathbb{N}_+}$. This is again a bounded sequence, so it has a convergent subsequence $\{x''_{n_{k_j}}\}_{j\in\mathbb{N}_+}$. Let us show that its limit must also be $\overline{x}$. First, we summarize all that we have established so far:

$$1)\ \lim_{n\to\infty} |x'_n - x''_n| = 0,$$
$$2)\ \lim_{k\to\infty} x'_{n_k} = \overline{x},$$
$$3)\ \lim_{j\to\infty} x''_{n_{k_j}} \text{ exists.}$$

So we have:

$$\begin{aligned}\lim_{j\to\infty}\left(x''_{n_{k_j}} - \overline{x}\right) &= \lim_{j\to\infty}\left(x''_{n_{k_j}} - x'_{n_{k_j}} + x'_{n_{k_j}} - \overline{x}\right)\\ &= \lim_{j\to\infty}\left(x''_{n_{k_j}} - x'_{n_{k_j}}\right) + \lim_{j\to\infty}\left(x'_{n_{k_j}} - \overline{x}\right) = 0\end{aligned}$$

So, indeed,

$$\lim_{j\to\infty} x''_{n_{k_j}} = \overline{x}.$$

The function f is continuous on I, so in particular it is continuous at $\overline{x} \in I$. This means that:

$$\lim_{j\to\infty} f(x'_{n_{k_j}}) = f(\lim_{j\to\infty} x'_{n_{k_j}}) = f(\overline{x}) = f(\lim_{j\to\infty} x''_{n_{k_j}}) = \lim_{j\to\infty} f(x''_{n_{k_j}}).$$

So we get that

$$\lim_{j\to\infty}\left(f(x'_{n_{k_j}}) - f(x''_{n_{k_j}})\right) = 0,$$

in contradiction to the assumption that

$$|f(x'_n) - f(x''_n)| \geq \varepsilon > 0, \qquad \forall n \in \mathbb{N}_+.$$

□

Example 9.14: The converse of Corollary 9.10 is false: the function $f(x) = \sqrt{x}$ is uniformly continuous on $[0,1]$ due to the Heine-Cantor theorem. However, as we saw in Example 9.10 it is not Lipschitz on $[0,1]$.